

A FAIR Journey

with the FAIRification Framework

eLwazi ODSP 2026 Data Jamboree

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www.elixir-europe.org



FAIRplus www.fairplus-project.eu

Outline

- The FAIR principles
- FAIRification journey
 - FAIRification Framework
 - How **FAIR** is your data? Define your FAIRification goals.
 - **Resources** to support your FAIRification journey
 - Some practical **considerations**

Reproducibility crisis: researchers' view



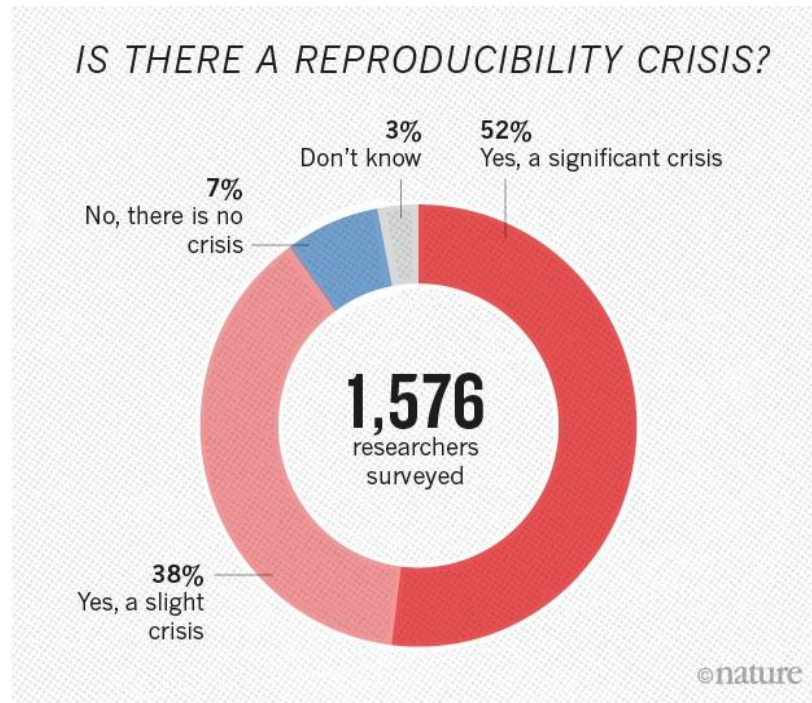
NATURE | NEWS FEATURE

1,500 scientists lift the lid on reproducibility

Survey sheds light on the 'crisis' rocking research.

Monya Baker

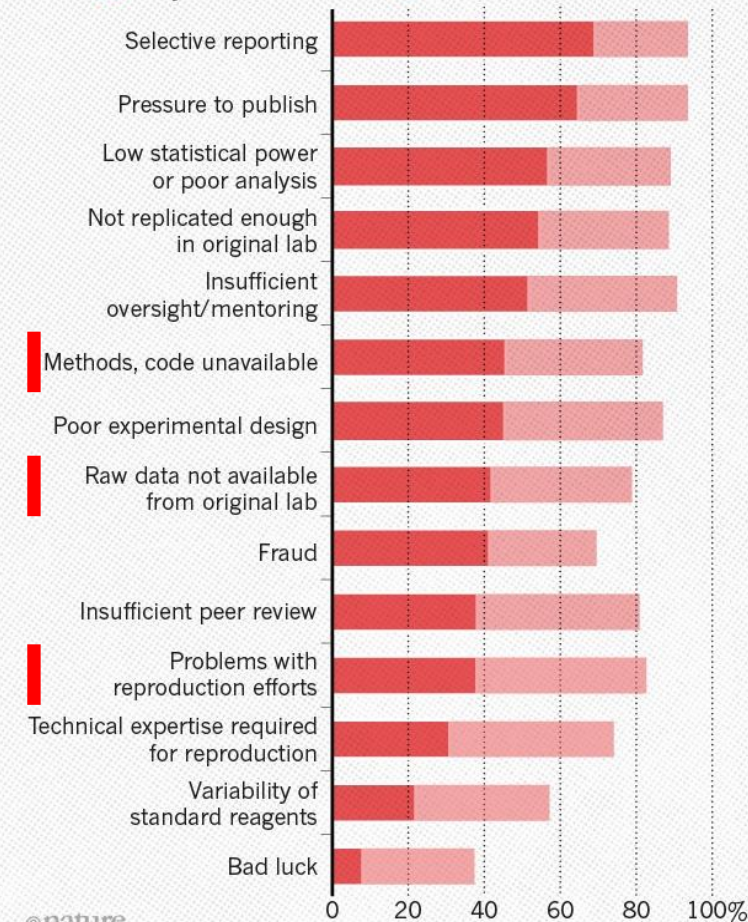
25 May 2016



WHAT FACTORS CONTRIBUTE TO IRREPRODUCIBLE RESEARCH?

Many top-rated factors relate to intense competition and time pressure.

● Always/often contribute ● Sometimes contribute



The FAIR guiding principles

The FAIR Guiding Principles for scientific data management and stewardship.
Wilkinson et al., Sci Data. 2016 Mar 15;3:160018. doi: 10.1038/sdata.2016.18.



Attention Score

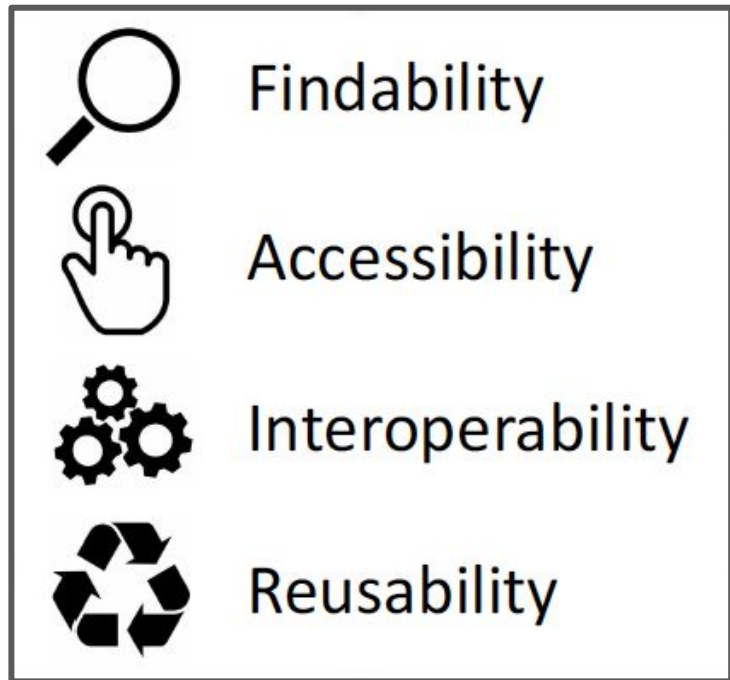
Access & Citations

1.12m

Article Accesses

15k

Citations



- ❑ The FAIR principles have to be considered as **inspiring** concepts. There are no strict rules.
- ❑ Diverse interpretation, ambiguity.
- ❑ Numerous resources, but no clear instructions on how to make data FAIR or to assess the data 'FAIRness'!

FAIR PRINCIPLES

- **To be Findable**
 - F1. (meta)data are assigned a globally unique and eternally persistent identifier.
 - F2. data are described with rich metadata.
 - F3. metadata include the data identifier it describes.
 - F4. (meta)data are registered or indexed in a searchable resource.
- **To be Accessible**
 - A1 (meta)data are retrievable by their identifier using a standardized communications protocol.
 - A1.1 the protocol is open, free, and universally implementable.
 - A1.2 the protocol allows for an authentication and authorization procedure, where necessary.
 - A2 metadata are accessible, even when the data are no longer available.
- **To be Interoperable**
 - I1. (meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
 - I2. (meta)data use vocabularies that follow FAIR principles.
 - I3. (meta)data include qualified references to other (meta)data.
- **To be Re-usable**
 - R1. meta(data) have a plurality of accurate and relevant attributes.
 - R1.1. (meta)data are released with a clear and accessible data usage license.
 - R1.2. (meta)data are associated with their provenance.
 - R1.3. (meta)data meet domain-relevant community standards.



The ultimate aim of
the FAIR principles is
Data Reuse!

THE FAIR PRINCIPLES



Can I find the data I want (unambiguously!) ?

➡ *Unambiguous identifiers supported by searchable metadata*

F1. (Meta)data are assigned a globally unique and persistent identifier

F2. Data are described with rich metadata (defined by R1 below)

F3. Metadata clearly and explicitly include the identifier of the data they describe

F4. (Meta)data are registered or indexed in a searchable resource

=> someone else could not reuse/reassign the same identifier without referring to your data

=> identifiers (links) should not become invalid over time

Examples

<http://www.uniprot.org/uniprot/P98161> (= > <https://www.uniprot.org/uniprotkb/P98161/entry>)

<https://doi.org/10.6084/m9.figshare.2056767.v2>

Globally unique and persistent identifiers are often provided by the repository or from a registry service.

THE FAIR PRINCIPLES



Can I find the data I want (unambiguously!) ?

➡ *Unambiguous identifiers supported by searchable metadata*

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F2. Data are described with rich metadata (defined by R1 below)

F3. Metadata clearly and explicitly include the identifier of the data they describe

F4. (Meta)data are registered or indexed in a searchable resource

=> "rich metadata" in F is for discovery, they describe what your dataset is about

Examples

Title, description, last version update, ...

Rich metadata in R is for reusability

THE FAIR PRINCIPLES



Can I find the data I want (unambiguously!) ?



Unambiguous identifiers supported by searchable metadata

F1. (Meta)data are assigned a globally unique and persistent identifier

F2. Data are described with rich metadata (defined by R1 below)

F3. Metadata clearly and explicitly include the identifier of the data they describe

F4. (Meta)data are registered or indexed in a searchable resource

=> the data identifier should be included in the metadata ...

Why?

Sometimes the metadata becomes separated from the data;

Thus, depending on how someone finds your data, it may be impossible to find a link back to the data or to the metadata ...

THE FAIR PRINCIPLES



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- ⇒ automated indexation by Google, Bing, ...
- ⇒ specific metadata indexes: data catalog (DCAT), ...

Examples

<https://www.fairdatapoint.org/>

IMI Data Catalog - <https://datacatalog.elixir-luxembourg.org/>

THE FAIR PRINCIPLES



Is there a way to access the data? Is it technically possible?

➡ *Clearly-defined access protocol, preferably machine-actionable*

A1. (Meta)data are **retrievable by their identifier using a standardised communications protocol**

A1.1 The protocol is open, free, and universally implementable

A1.2 The protocol allows for an authentication and authorisation procedure where necessary

A2. Metadata are accessible, even when the data are no longer available

- ⇒ focuses on how data and metadata can be retrieved (not “resolved”; A1 does not require web!)
- ⇒ defines what the protocol is to request access to the data (to be compliant)

THE FAIR PRINCIPLES



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- ⇒ whatever the protocol, one should not have to buy a licence to use the tools required to access the data

Examples

https, sFTP ...

Telephone! (yes: the protocol may involve human intervention)

THE FAIR PRINCIPLES



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Examples

https, SFTP ...

Telephone! (yes: the protocol may involve human intervention)

⇒ open protocol, not open data!

⇒ authorization via telephone, email, web, ...

FAIR ≠ Open Data

THE FAIR PRINCIPLES



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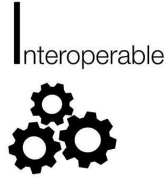
A2. Metadata are accessible, even when the data are no longer available

- ⇒ It is therefore recommended that data and metadata are separated!
- 1 database for the metadata and 1 database for the data;
 - do not embed all the metadata in the images;

Why?

- ⇒ metadata are more discoverable by metadata aggregators, such as OpenAIR <https://www.openaire.eu/>
- ⇒ metadata are valuable even if the data is lost or not available anymore

THE FAIR PRINCIPLES



Can I understand the data? or better, can my machine understand the data?



Use shared vocabularies/ontologies in machine-accessible format

11. (Meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
12. (Meta)data use vocabularies that follow FAIR principles
13. (Meta)data include qualified references to other (meta)data

⇒ most widely used for FAIRfication projects: Resource Description Framework (RDF)

Why?

Standard specifically designed to describe any type of information in a structured and machine-readable format.

Alternatives:

- JSON-LD – JavaScript Object Notation for Linking Data – <https://json-ld.org/>
- Microdata – <https://en.wikipedia.org/wiki/Microdata> (HTML)
- ...

THE FAIR PRINCIPLES

Interoperable



Can I understand the data? or better, can my machine understand the data?

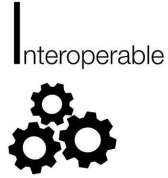


Use shared vocabularies/ontologies in machine-accessible format

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- I3. (Meta)data include qualified references to other (meta)data

⇒ the vocabularies describing the (meta)data must themselves be Findable, Accessible, Interoperable and Reusable

THE FAIR PRINCIPLES



Interoperable

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qualified references

⇒ cross-references that explains its intent: "... is a regulator of ..."



P04278 · SHBG_HUMAN

Protein ⁱ	Sex hormone-binding globulin
Gene ⁱ	SHBG
Status ⁱ	UniProtKB reviewed (Swiss-Prot)
Organism ⁱ	Homo sapiens (Human)

Are these good qualified references?

For humans, yes. But not for machines ...

```
SELECT ?protein ?organism
WHERE
{
    ?protein a up:Protein .
    ?protein up:organism taxon:9606 .
}
```

Proteins

derived
from

Homo
sapiens

THE FAIR PRINCIPLES

R_{eusable}



Am I allowed to use the data? SHOULD I re-use the data? Is it useful for me?



Contextual information, allowing proper interpretation



Rich provenance information facilitating accurate citation

R1. (Meta)data are richly described with a plurality of accurate and relevant attributes

R1.1. (Meta)data are released with a clear and accessible data usage license

R1.2. (Meta)data are associated with detailed provenance

R1.3. (Meta)data meet domain-relevant community standards

⇒ focuses on the ability of a user (machine or human) to decide if the data is actually
USEFUL

⇒ metadata richly describes the context under which the data was generated
plurality : do as much as you can; be generous;

Examples

experimental protocol

the manufacturer and brand of the machine or sensor that created the data

the species used

the drug regime

the particularities, the limitations

...

Rich metadata in F is for discovery

THE FAIR PRINCIPLES

R _{reusable}



Am I allowed to use the data? SHOULD I re-use the data? Is it useful for me?



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⇒ data usage license: bad wording

(license or data access policy would have been better)

The conditions under which the data can be used should be clear to machines and humans.

THE FAIR PRINCIPLES

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⇒ they should allow the user to decide whether or not they **trust** the data

what did you do?

what organism was used?

what protocols were followed?

what was the question?

which workflows, software and versions of, filters?

who and how to cite?

etc.

THE FAIR PRINCIPLES

R _{reusable}



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R1.3. (Meta)data meet domain-relevant community standards

⇒ "catch-all"

⇒ in addition to all the previous principles, you need to adhere to community standards

you should use your community standard ...

... even if it's not FAIR! And then you should also use a FAIR standard

FAIR PRINCIPLES

F_{indable}



Can I find the data I want (unambiguously!) ?

➡ Unambiguous identifiers supported by searchable metadata

A_{ccessible}



Is there a way to access the data? CAN I re-use the data? Is it technically possible?

➡ Clearly-defined access protocol, preferably machine-actionable

I_{nteroperable}



Can I understand the data? or better, can my machine understand the data?

➡ Use shared vocabularies/ontologies in machine-accessible format

R_{eusable}



Am I allowed to use the data? SHOULD I re-use the data? Is it useful to my problem?

➡ Contextual information, allowing proper interpretation
➡ Rich provenance information facilitating accurate citation

FAIRification journey – the FAIRification Framework (3 key components)

1. The FAIRification Process

is the reusable process which outlines the 4 main phases of a FAIRification activity.

1. FAIRification Process



Phase 1

set realistic and practical goals – identify the intended usability of the data that cannot be achieved in its current state.

FAIRification Process – the FAIRification Framework

1. The FAIRification Process

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1. FAIRification Process



Phase 2

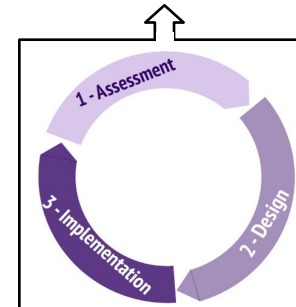
- Identify Data Requirements
- Identify FAIRification Capabilities & Resources
- Produce the FAIRification Backlog

FAIRification Process – the FAIRification Framework

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1. FAIRification Process



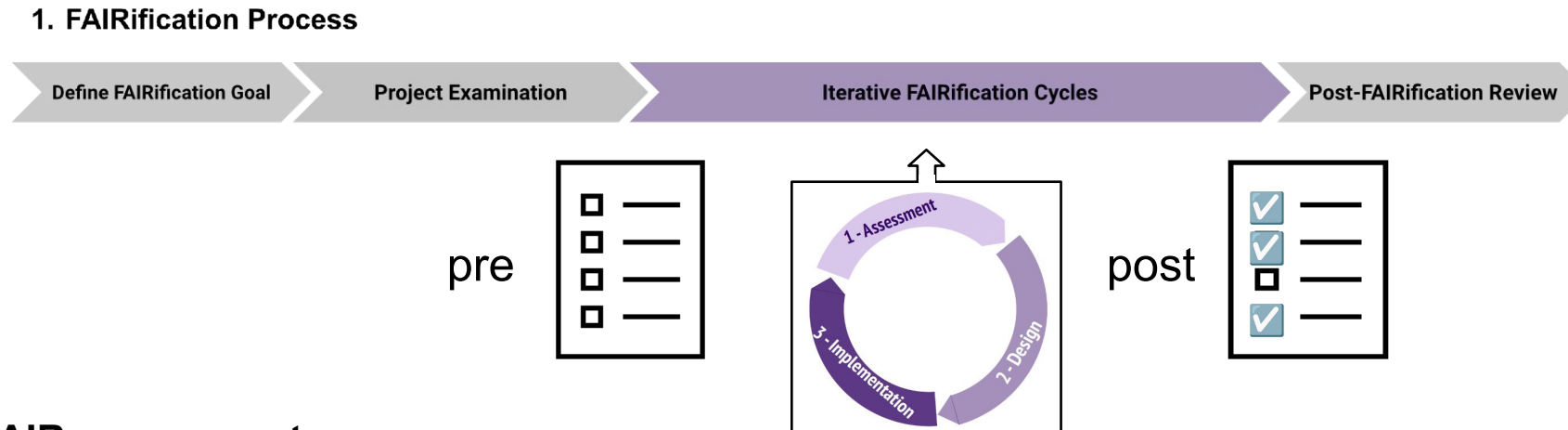
Phase 3

1. Pre- and post-FAIR assessment
2. Identify concrete steps / tasks (FAIRification template)
3. Execute planned tasks within the established timeframe.

FAIRification Process – the FAIRification Framework

1. The FAIRification Process

is the reusable process which outlines the 4 main phases of a FAIRification activity.



FAIR assessment

- [Dataset Maturity Model](#) (DSM) defines and classifies requirements that constitute an incremental path towards improving FAIRness level for a given research dataset.
- Other FAIR assessment resources: <https://fairassist.org/tools> (often a check-list of the FAIR principles)

FAIRification Process – the FAIRification Framework

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1. FAIRification Process



Phase 4

Peer review of all the outputs of the FAIRification processes against the initial goals to assess the overall success.

FAIRification Process – the FAIRification Framework

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1. FAIRification Process



FAIRification journey – the FAIRification Framework (3 key components)

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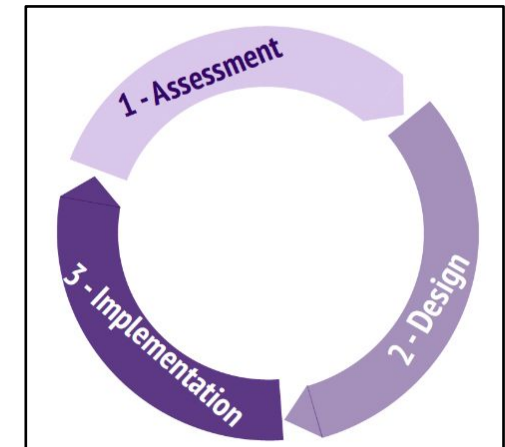
2. The FAIRification Template

proposes a set of 8 clear & distinct steps for the implementation stage (phase 3 – Iterative FAIRification Cycles / Design)

1. FAIRification Process



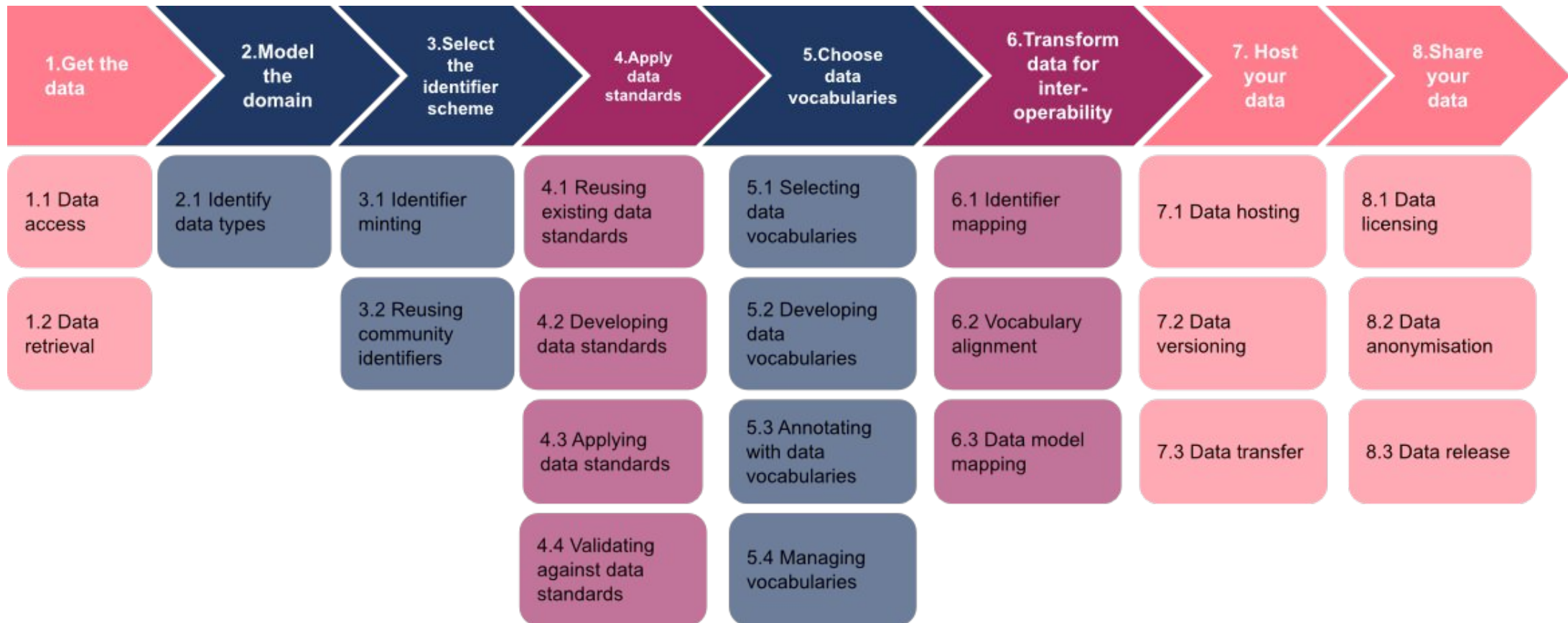
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FAIRification Framework – the FAIRification Template

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FAIRification journey – the FAIRification Framework (3 key components)

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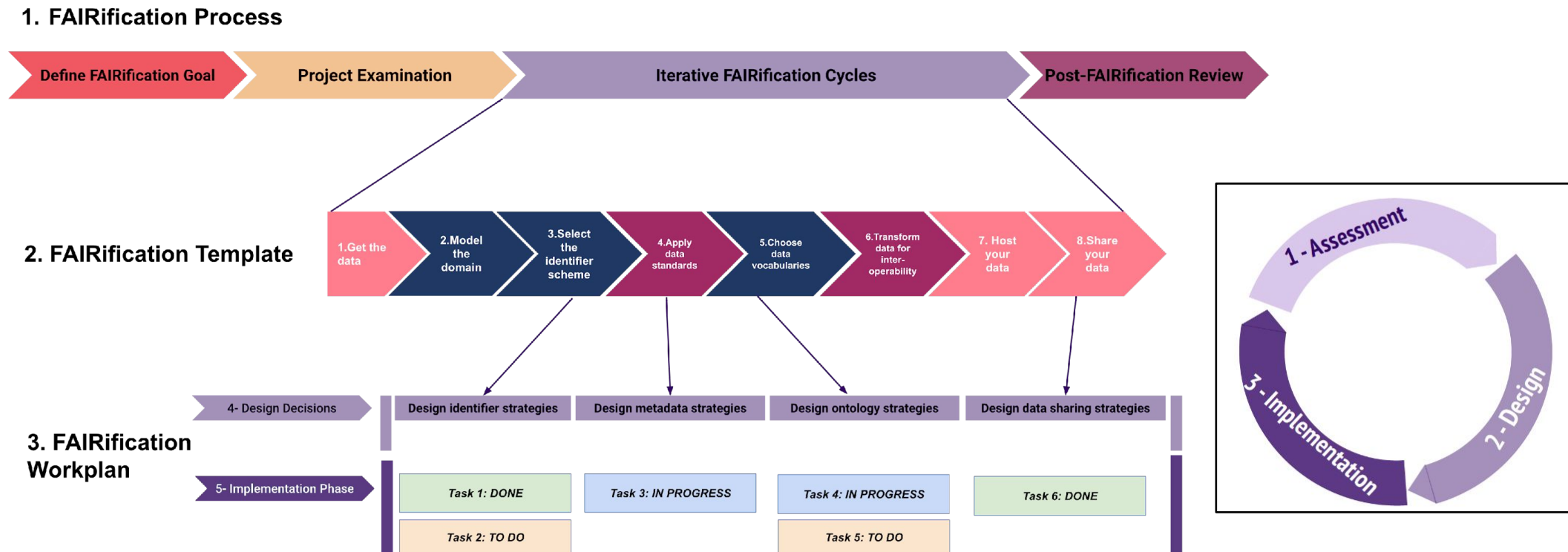
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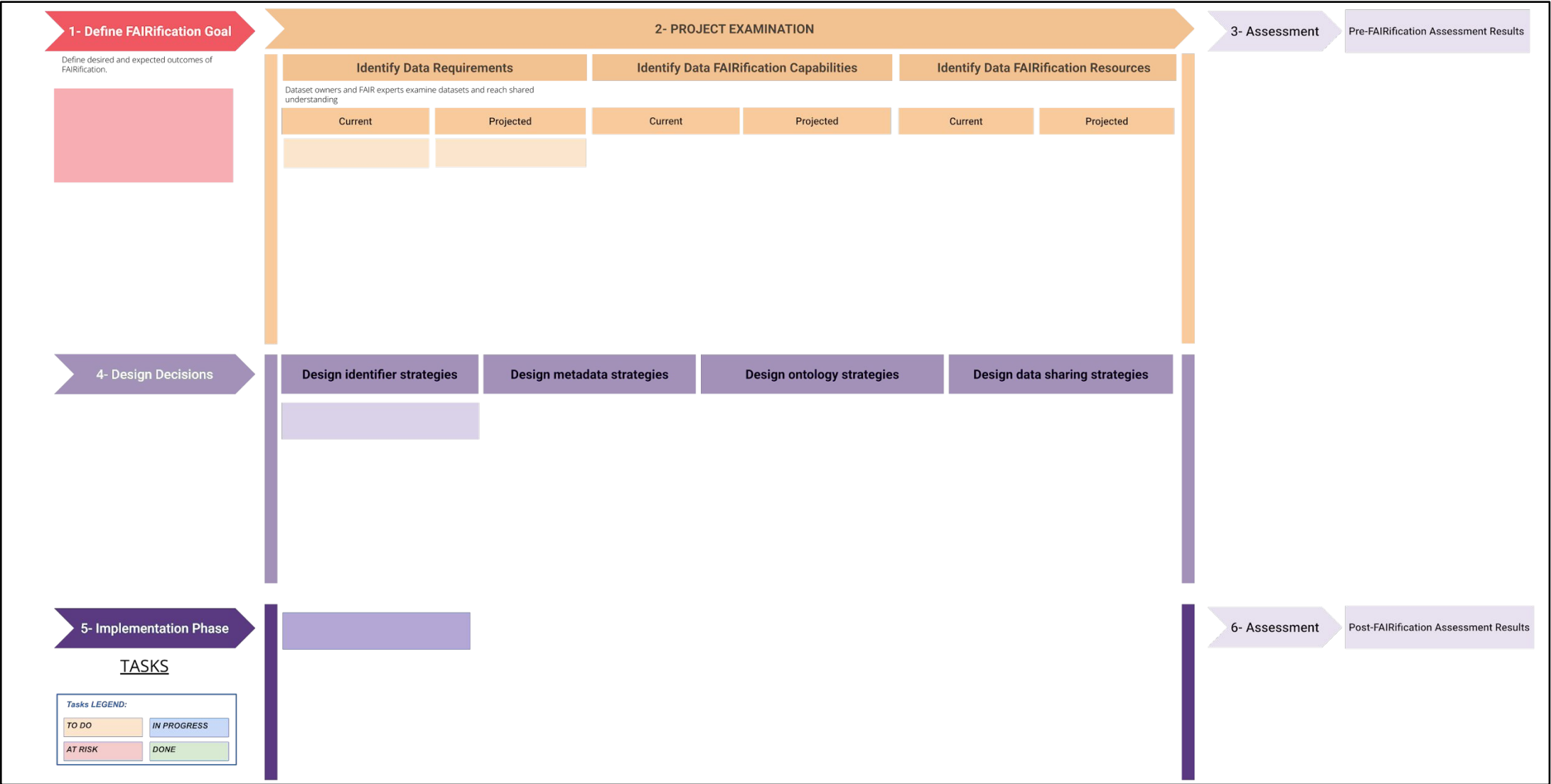
3. The FAIRification Workplan

provides a structure for organizing FAIR implementation work tailored to the needs of a specific project or use case.



FAIRification Framework – the FAIRification Template

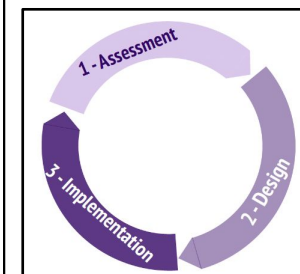
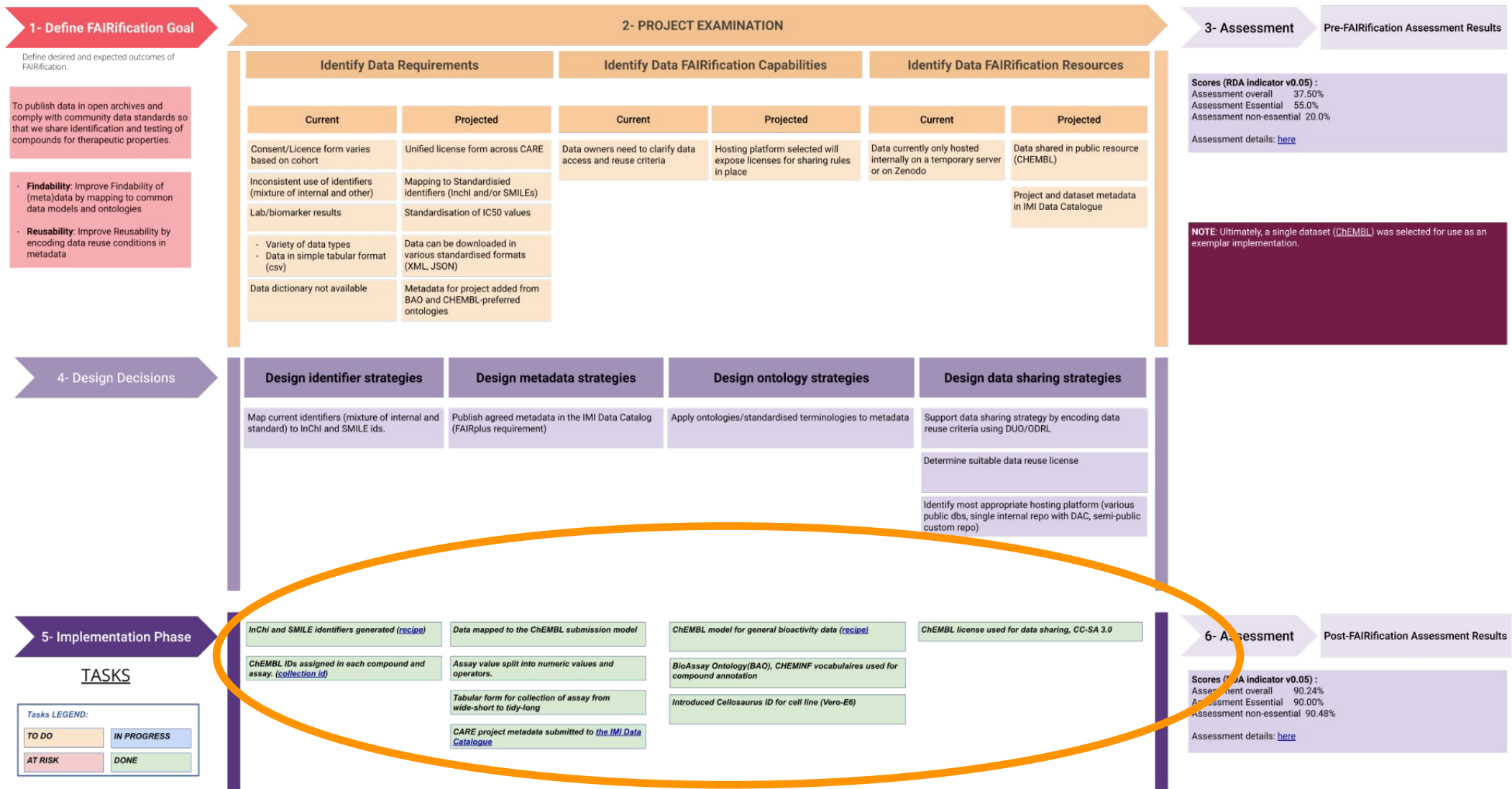
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FAIRification Framework – the FAIRification Template

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HOW FAIR IS YOUR DATA?



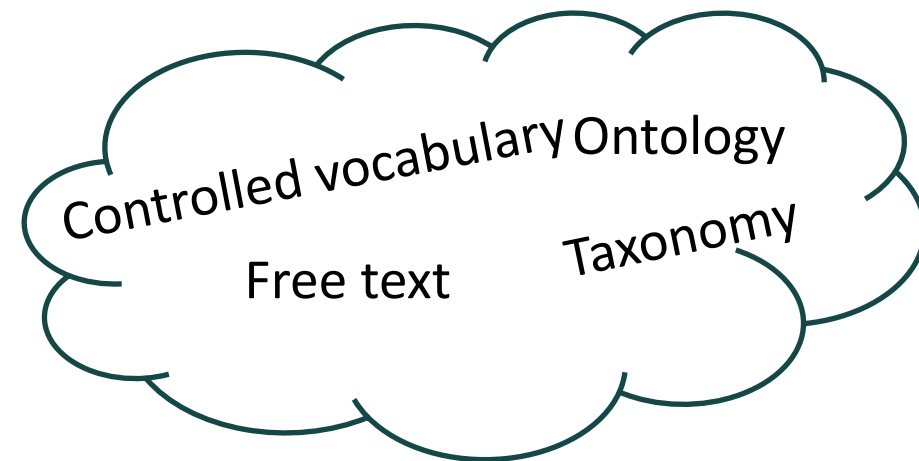
FA ≠ FAIR

- Very Good **FA** is still ≠ **FAIR**
- (Re)-**Use** is the goal

NOT Findability, NOT Accessibility and NOT Interoperability

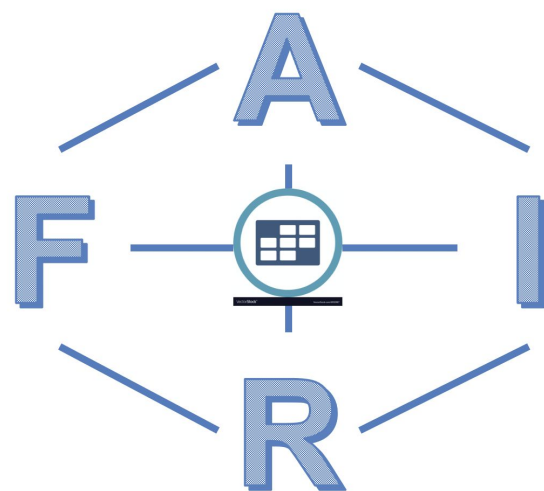
The ultimate aim
of the FAIR
principles is
Data Reuse!

- Which would you use?
- How would you rank them and why?



HOW FAIR IS YOUR DATA?

Assessing FAIR as a **state** vs FAIR as an **Alphabet**



State of FAIRness

vs

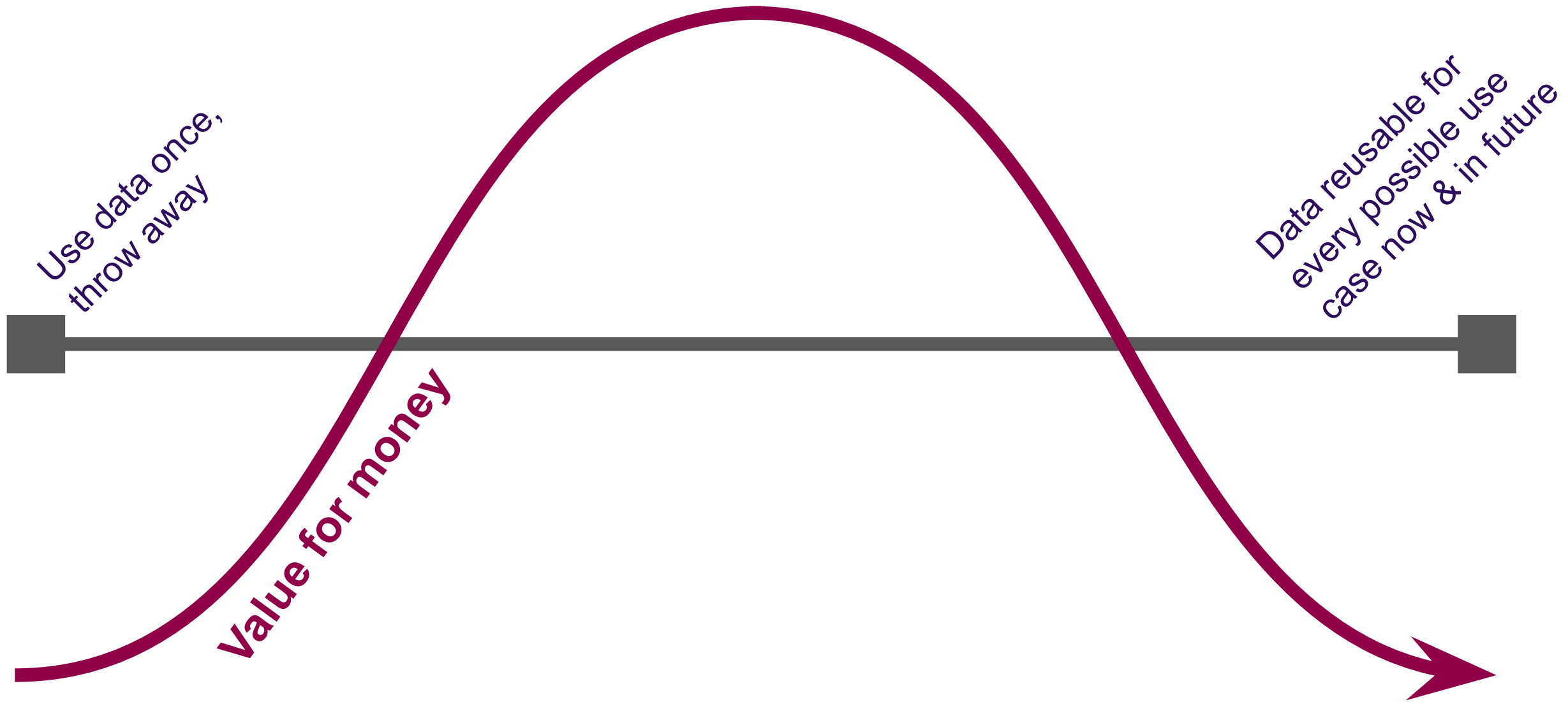


FAIR score

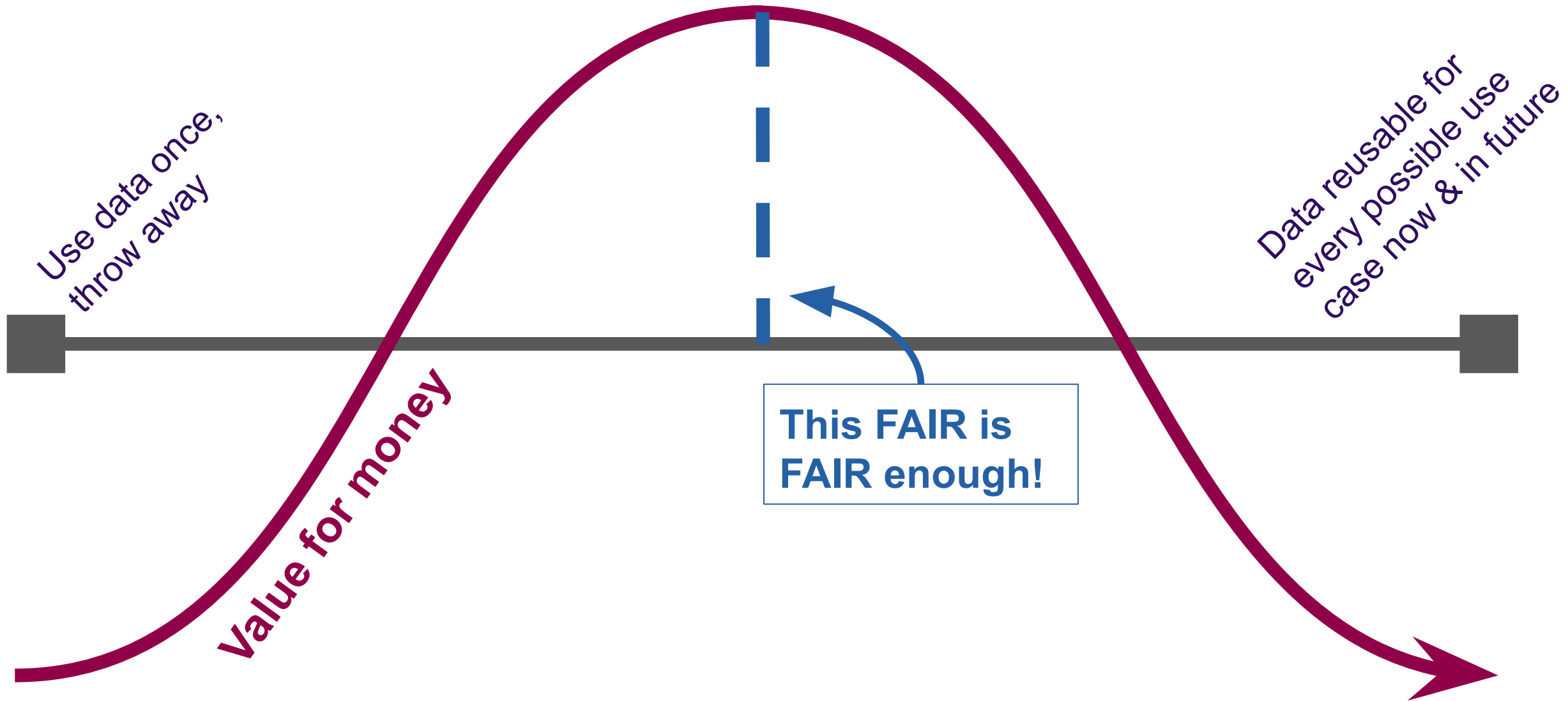
HOW FAIR IS YOUR DATA? Is it FAIR enough?



HOW FAIR IS YOUR DATA? Is it FAIR enough?



HOW FAIR IS YOUR DATA? Is it FAIR enough?



| What makes a good FAIRification goal?

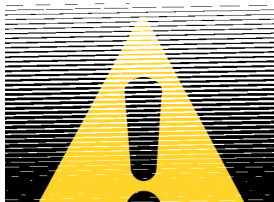


WHAT: be **SPECIFIC** about the FAIRification task you want to accomplish

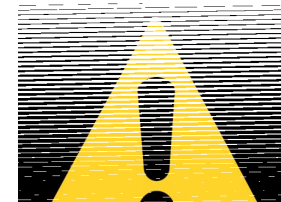
WHY: define a **CLEAR SCIENTIFIC SCOPE** for the goal

HOW: ensure that your goal is **ACTIONABLE**

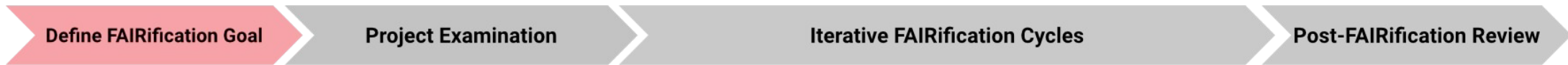
END STATE: before you start, define what **FAIR ENOUGH** looks like for your dataset



Do **NOT** use the word **FAIR!!!**



An example of a good FAIRification goal



WHAT: To make the project's bioactivity data comply with community standards and publicly available

WHY: so that other researchers can easily reuse the data without repeating the compound identification and testing work

HOW: To make the project's bioactivity data comply with community standards and publicly available* so that other researchers can easily reuse the data without repeating the compound identification and testing work

**could be even more specific and actionable, e.g. by specifying which community standard and publicly available from where*

FAIRification goals activity from previous data jamboree



Outline

- The FAIR principles
- FAIRification journey
 - How **FAIR** is your data? Define your FAIRification goals.
 - *FAIRification Framework*
 - **Resources** to support your FAIRification journey
 - Some practical **considerations**

2

Resources to support your FAIRification journey



FAIRplus Tailored FAIRification Process - CARE - Iteration 2

1- Define FAIRification Goal

Define desired and expected outcomes of FAIRification.

To publish data in open archives and comply with community data standards so that we share identification and testing of compounds for therapeutic properties.

- **Findability:** Improve Findability of (meta)data by mapping to common data models and ontologies
- **Reusability:** Improve Reusability by encoding data reuse conditions in metadata

2- PROJECT EXAMINATION

Identify Data Requirements

Current	Projected
Consent/License form varies based on cohort	Unified license form across CARE
Inconsistent use of identifiers (mixture of internal and other)	Mapping to Standardised identifiers (Inchi and/or SMILES)
Lab/biomarker results	Standardisation of IC50 values
- Variety of data types - Data in simple tabular format (csv)	Data can be downloaded in various standardised formats (XML, JSON)
Data dictionary not available	Metadata for project added from BAO and ChEMBL vocabularies

Identify Data FAIRification Capabilities

Current	Projected
Data owners need to clarify data access and reuse criteria	Hosting platform selected will expose licenses for sharing rules in place

Identify Data FAIRification Resources

Current	Projected
Data currently only hosted internally on a temporary server or on Zenodo	Data shared in public resource (ChEMBL)
	Project and dataset metadata in IMI Data Catalogue

Design Decisions

Design identifier strategies

Map current identifiers (mixture of internal and standard) to InChi and SMILE ids.

Design metadata strategies

Publish agreed metadata in the IMI Data Catalog (FAIRplus requirement)

Design ontology strategies

Apply ontologies/standardised terminologies to metadata

Design data sharing strategies

Support data sharing strategy by encoding data reuse criteria using DUO/ODRL

Determine suitable data reuse license

Identify most appropriate hosting platform (various public dbs, single internal repo with DAC, semi-public custom repo)

5- Implementation Phase

TASKS

Tasks LEGEND:

TO DO	IN PROGRESS
AT RISK	

InChi and SMILE identifiers generated ([recipio](#))

ChEMBL IDs assigned in each compound and assay ([collection id](#))

Data mapped to the ChEMBL submission model

Assay value split into numeric values and operators.

Tabular form for collection of assay from wide-short to tidy-long

CARE project metadata submitted to [the IMI Data Catalogue](#)

ChEMBL model for general bioactivity data ([recipio](#))

BioAssay Ontology(BAO), ChEMIN vocabularies used for compound annotation

Introduced Cellosaurus ID for cell line (Vero-E6)

ChEMBL license used for data sharing, CC-SA 3.0

3- Assessment

Pre-FAIRification Assessment Results

Scores (RDA indicator v0.05) :

Assessment overall	37.50%
Assessment Essential	55.0%
Assessment non-essential	20.0%

Assessment details: [here](#)

NOTE: Ultimately, a single dataset (ChEMBL) was selected for use as an exemplar implementation.

6- Assessment

Post-FAIRification Assessment Results

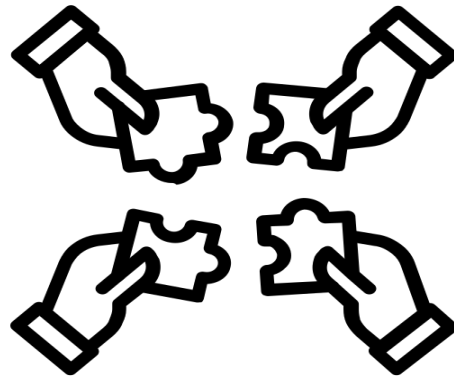
Scores (RDA indicator v0.05) :

Assessment overall	90.24%
Assessment Essential	90.00%
Assessment non-essential	90.48%

Assessment details: [here](#)

You are looking for...

Guidance about FAIR and Data Management Planning



The FAIR Cookbook



FAIRCOOKBOOK [GITHUB](#) Search Wizard...

F Findability
 EXEMPLAR RECIPES

- Unique, persistent identifiers
- Search engine optimization

 LEARN MORE

A Accessibility
 EXEMPLAR RECIPES

- Transferring data with SFTP
- Downloading data with Aspera

 LEARN MORE

I Interoperability
 EXEMPLAR RECIPES

- Selecting terminologies and ontologies
- Creating a metadata profile

 LEARN MORE

R Reusability
 EXEMPLAR RECIPES

- Data licenses
- Declaring data's permitted uses

 LEARN MORE

Infrastructure
 LEARN MORE

Assessments
 LEARN MORE

Applied Examples
 LEARN MORE

Maturity model
 LEARN MORE

↓

← FAIRCOOKBOOK [GITHUB](#) Search Wizard...

3. Identifier mapping with BridgeDb

Contents

- 3.1. Main Objectives
- 3.2. Graphical Overview
- 3.3. Requirements
- 3.4. Tools
- 3.5. Identifier mapping with BridgeDb
- 3.6. Provenance
- 3.7. Code
- 3.8. Conclusion
- 3.9. References
- 3.10. Authors
- 3.11. License

Recipe Overview

- Reading Time: 30 minutes
- Executable Code: No
- Difficulty: 4/5

Mapping identifiers with BridgeDb

Recipe Type: Hands-on

Audience: Principal Investigator, Data Manager, Data Scientist

Maturity Level & Indicator: DSM-4-C4

Cite me with FCB017

3.1. Main Objectives

The main purpose of this recipe is to:

Provide practical examples on how to **map identifiers for Genes, Proteins, Metabolites and Pathways** between resources using a purpose built tool, namely **BridgeDb**. Hands on guidance is provided for 2 interfaces (R package and a Python Webservices) provided by **BridgeDb**.

The FAIR Cookbook

What is it?

An online, 'live' resource for the **life sciences**

A collection of **recipes** that cover the operation steps of FAIR data management

Who is it for?

Anyone who needs practical assistance in their FAIRification journey or creates FAIR guidance and educational material

Who developed it?

Researchers and **data managers professionals** in the life sciences, from academia and industry
Including ELIXIR members

<https://faircookbook.elixir-europe.org>

Found a coverage gap? Want to share your expertise?

What to contribute?

- Contribute test datasets
- Contribute examples
- Contribute code

Why participating?

- Get exposure
- Get cited
- Get feedback

Topics of interest:

- ETL to RDF solutions
- KG generation from unstructured data
- GDPR compliant document
- Input validation & RDF shape constraint validation
- Specific FAIRification process

The Research Data Management toolkit for Life Sciences



The RDMkit logo, consisting of the text 'RDMkit' in a dark blue and magenta font.

[Data management](#) [About](#) [Contribute](#) [G](#)

The Research Data Management toolkit for Life Sciences
Best practices and guidelines to help you make your data FAIR (Findable, Accessible, Interoperable and Reusable)

What can we help you find?

Search RDMkit

Browse all topics by

Data life cycle
Start here to get an overview of research data management based on stages in the data life cycle.

Your role
Identify your role in research data management, find data management resources relevant for you, and information to help you progress in your career path.

Your domain
Learn about data management tasks that affect your domain or research community, and the solutions adopted to address them.

Data management

- Data life cycle >
- Your role >
- Your domain >**
- Your tasks v
 - Compliance monitoring
 - Costs of data management
 - Creating a data-flow diagram
 - Data analysis
 - Data brokering
 - Data deletion
 - Data discoverability
 - Data interlinking
 - Data management coordination
 - Data management plan
 - Data organisation
 - Data security
 - Data sensitivity
 - Data provenance



<https://rdmkit.elixir-europe.org/>

Data Stewardship Wizard



DS Wizard

Users

Knowledge Models

Projects

List

Importers

Documents

Settings

Albert Einstein

Admin

« Collapse sidebar

My Experiment

Questionnaire

Metrics

Preview

Documents

Settings

View

Import answers

Comments 4

TODOs 1

Version history

Named versions only

October 2022

4. 10.

9:38

Current 1.0.0

Will this work space be run by dedicated specialists?

Yes

Albert Einstein

9:38

Will you be using a shared working space to work with your data?

Yes

Albert Einstein

9:37

Data format/type

XML Query Language

Albert Einstein

9:34

Cleared reply of Data set:

Current Phase

Before Submitting the Proposal

Chapters

I. Administrative information

II. Re-using data

III. Creating and collecting data

IV. Processing data

V. Interpreting data

VI. Preserving data

VII. Giving access to data

IV. Processing data

In the processing phase, the data will be undergoing the mostly automated steps for processing, before the analysis and interpretation.

In this chapter, many questions are focusing on the compute environment that is used to process the data and make it available for interpretation by project partners. Some of those questions (e.g. on workflow systems and data provenance) are also relevant for the work in the interpretation phase.

1 Will you be using a shared working space to work with your data?

Horizon 2020 DMP

Horizon Europe DMP

Science Europe DMP

Will you be using a working space containing data and software specific to the project that is shared between all the people working on the data in the project? Sometimes such a system is called a Virtual Research Environment.

Desirable: Before Submitting the Proposal

a. No, participants in the project each have different collections of data and tools

b. Yes

Clear answer

Answered 5 minutes ago by Albert Einstein.

1.b.1 Will this work space be run by dedicated specialists?

TODO

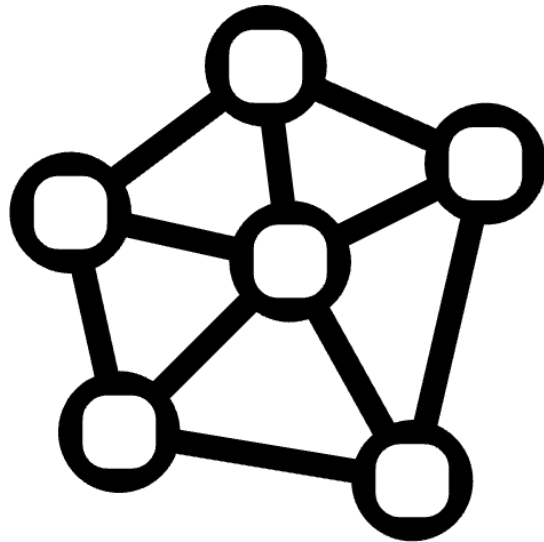
If your work space is run and maintained by specialists, e.g. the ICT department of one of the institutes involved in the projects, this means that backup and restore as well as access management is properly addressed.

Desirable: Before Submitting the DMP



You are looking for...

Standards, ontologies & vocabularies



FAIRsharing



The screenshot shows the top of the FAIRsharing.org website. At the top left is the logo. To its right is a search bar with the placeholder text 'search through all content'. Further right are two buttons: 'SEARCH' and 'ADVANCED SEARCH'. On the far right is a 'LOGIN' button with a right-pointing arrow. Below these are seven navigation buttons: 'STANDARDS', 'DATABASES', 'POLICIES', 'COLLECTIONS', 'ORGANISATIONS', 'ADD CONTENT', and 'STATS'. Below the navigation bar is a large dark blue hero section with white text. The main headline reads: 'A curated, informative and educational resource on data and metadata standards, inter-related to databases and data policies.' Below this are three lines of text in a lighter blue color: 'Guides consumers to discover, select and use these resources with confidence.', 'Helps producers to make their resources more visible, more widely adopted and cited.', and 'Provides humans and tools with access to trustworthy content to enable data management tasks.'

FAIRsharing.org standards, databases, policies

search through all content

SEARCH ADVANCED SEARCH LOGIN →

STANDARDS DATABASES POLICIES COLLECTIONS ORGANISATIONS ADD CONTENT STATS

A curated, informative and educational resource on data and metadata standards, inter-related to databases and data policies.

Guides **consumers** to discover, select and use these resources with confidence.

Helps **producers** to make their resources more visible, more widely adopted and cited.

Provides **humans** and **tools** with access to trustworthy content to enable data management tasks.



The screenshot shows the 'FAIRsharing Assistant Beta' interface. At the top is a dark blue header with the text 'FAIRsharing Assistant Beta' in white. Below the header is the FAIRsharing.org logo. The main content area has a light blue background. It contains the text: 'FAIRsharing is a curated informative and educational resource on data and metadata standards, inter-related to databases and data policies.' followed by 'This Assistant will help you find the standards, databases and policies you need to help make your data FAIR.' At the bottom center is a blue button with the text 'GET STARTED'.

FAIRsharing Assistant *Beta*

FAIRsharing.org standards, databases, policies

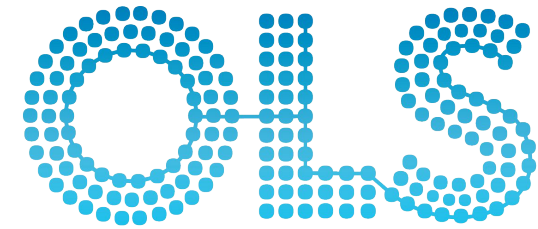
FAIRsharing is a curated informative and educational resource on data and metadata standards, inter-related to databases and data policies.

This Assistant will help you find the standards, databases and policies you need to help make your data FAIR.

GET STARTED

<https://assist.fairsharing.org/>

Ontology Lookup Service





Home | Ontologies | Help | About | Downloads

Search

☐ Exact match ☐ Include obsolete terms ☒ Include imported terms

Showing 10 from a total of 2287

Type

- ☐ Class (2283)
- ☐ Individual (2)
- ☐ Objectproperty (2)
- ☐ Property (2)

Ontology

- ☐ AERO (1)
- ☐ AGRO (1)
- ☐ APOLLO_SV (1)
- ☐ BAO (11)
- ☐ CCO (155)
- ☐ CHEMINF (1)
- ☐ CHIRO (3)
- ☐ CIDO (6)
- ☐ CPONT (15)
- ☐ CVDO (22)
- ☐ DIDEO (1)
- ☐ DOID (142)
- ☐ DUO (1)
- ☐ EFO (241)
- ☐ ENM (18)
- ☐ EPIO (3)
- ☐ GENEPIO (5)
- ☐ GEXO (153)
- ☐ HCAO (1)
- ☐ HP (12)

Search results for: cardiomyopathy

Previous 1 2 ... 229 Next

cardiomyopathy EFO:0000318

http://www.ebi.ac.uk/efo/EFO_0000318

A disease of the heart muscle or myocardium proper. Cardiomyopathies may be classified as either primary or secondary, on the basis of etiology, or on the pathophysiology of the lesion: hypertrophic, dilated, or restrictive.

Ontology: EFO

Also appears in: PRIDE

cardiomyopathy MONDO:0004994

http://purl.obolibrary.org/obo/MONDO_0004994

A disease of the heart muscle or myocardium proper. Cardiomyopathies may be classified as either primary or secondary, on the basis of etiology, or on the pathophysiology of the lesion: hypertrophic, dilated, or restrictive.

Ontology: MONDO

Also appears in: CPONT OBA

cardiomyopathy DOID:0050700

http://purl.obolibrary.org/obo/DOID_0050700

A heart disease and a myopathy that is characterized by deterioration of the function of the heart muscle. MESH:D009202 added from NeuroDevNet [WAK].

Ontology: DOID

Also appears in: ONTONEO HTN BAO EPIO GENEPIO CVDO RBO

cardiomyopathy MPATH:615

http://purl.obolibrary.org/obo/MPATH_615

A disease of the heart muscle or myocardium proper. Cardiomyopathies may be classified as either primary or secondary, on the basis of etiology, or on the pathophysiology of the lesion: hypertrophic, dilated, or restrictive.

Ontology: MPATH

Also appears in: MP UPHENO

cardiomyopathy MP:0005330

http://purl.obolibrary.org/obo/MP_0005330

diseases of the heart (myocardium); may result from many causes

Ontology: MP

Also appears in: CHIRO UPHENO



<https://www.ebi.ac.uk/ols4/>

ISA Framework



 isatools Software Suite ▾ Model and Formats ▾ Blog Support User Community and Publications						
Configuration	Data Annotation	Programmable APIs	ISA-Tab Parsers	Conversion	Web	Analysis
 ISAcreator configurator	 ISAcceptor	 Python API	 PERL Parser	 ISAconverter	 BioInvestigation Index	 RISA
	 OntoMaton		 Python Parser	 MageToISA	 ISA-Tab Viewer	 ISA Galaxy
	 NanoMaton			 LinkedISA		
				 ISAtorDF		

<https://isa-tools.org/>



You are looking for...

Identifiers & Identifier Matching



Identifiers.org



Identifiers.org
Resolution service

 **Resolution**

 Registry

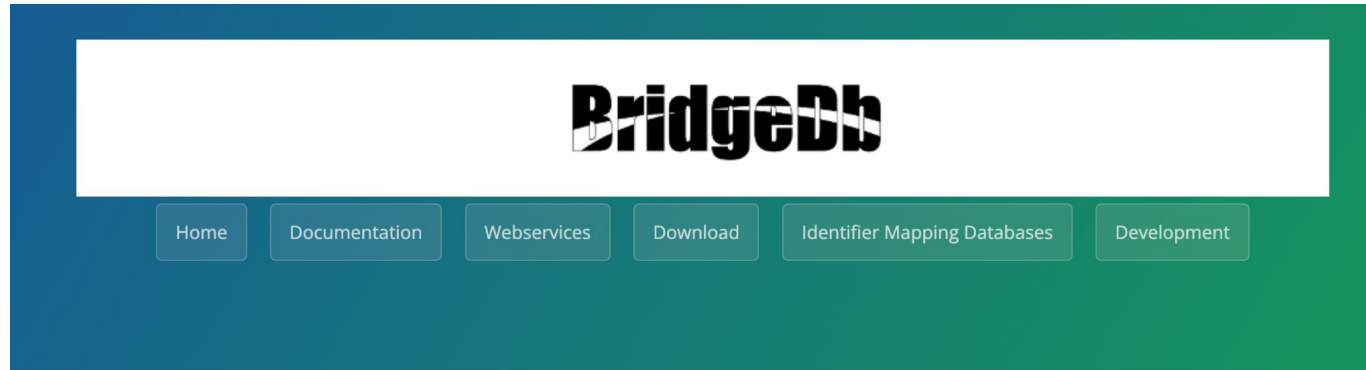
 Browse the registry

 Make a request ▾

 Documentation



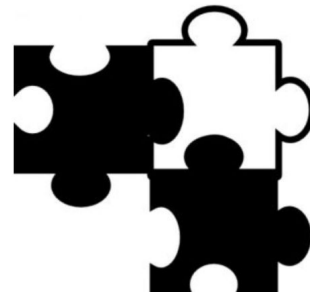
BridgeDB



Standardised access to Identifier Mappings

BridgeDb is a framework to map identifiers between various biological databases and related resources. These mappings are provided for genes, proteins, metabolites, metabolic reactions, diseases, complexes and publications. BridgeDb includes a Java library that provides an API for programmatic access.

<https://www.bridgedb.org/>



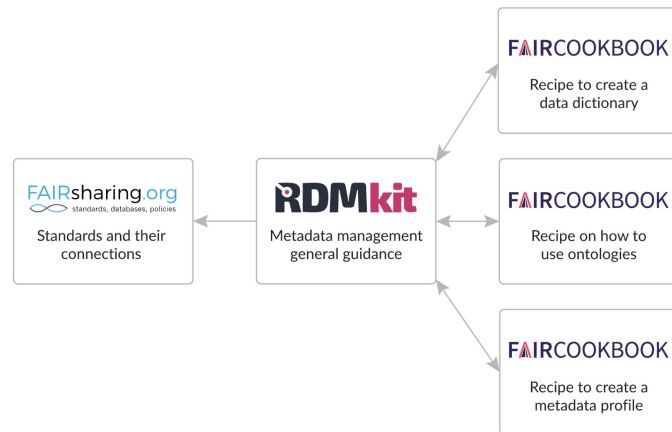
The ELIXIR FAIR RDM know-how ecosystem

ELIXIR RDM guidelines

How can I find relevant metadata standards for FAIR metadata handling?

Audience: Data stewards, researchers, PIs, research software engineers..

- The [documentation and metadata page in RDMkit](#) will help you to understand what you need to consider when managing the metadata for your project. To further explore the standards used in your domain as well as their connections, a link to FAIRsharing is provided.
- Several relevant FAIR Cookbook recipes are linked from RDMkit that can help with the task. For example, you can follow instructions to [create a data dictionary](#) to define the metadata variables you need to capture - this recipe is accessible to most researchers and does not require a great level of technical knowledge.
- If you want to go further, check out the recipe on [how to use ontologies](#) to describe your variables and values - this recipe requires some understanding on what ontologies are and how to use them. If you want to go beyond metadata standards and data dictionaries, a further recipe guides you through [creating a metadata profile](#) from scratch, if no existing standard fully meets your needs. The FAIR Cookbook recipes also link you back to RDMkit for broader context.



<https://elixir-europe.org/platforms/interoperability/rirs>



<https://bio.tools/>



<https://tess.elixir-europe.org/>

<https://elixir-europe.org/what-we-offer/guidelines/data-management>



<https://workflowhub.eu/>

Repositories and databases

- ELIXIR Core Data Resources
<https://elixir-europe.org/platforms/data/core-data-resources>
- ELIXIR Deposition Databases for Biomolecular Data
<https://elixir-europe.org/platforms/data/elixir-deposition-databases>
- Global Biodata Coalition Core Biodata Resources
<https://globalbiodata.org/what-we-do/global-core-biodata-resources/list-of-current-global-core-biodata-resources/>

3 Some practical considerations

Financial

Costs - time, data infrastructure, personnel, ROI, ...

Technical

Infrastructure, tools, methodologies, ...

Legal

Accessibility rights – "Accessible", "Reusable" including data FAIRification

Organisational

Training, development of an organisation culture